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Joshua A. Brown

More
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Summary

Staff software engineer with 12 years of experience building enterprise software, developer platforms, and AI products. Hands-on full-stack technical leader spanning cross-team delivery, reusable architecture, and mentoring, with work ranging from cross-device Copilot experiences to web interfaces serving 70 million daily views.

Technical Skills

Languages:	TypeScript, JavaScript, Go, Python, Ruby, C#, C++, SQL
Front end:	React, HTML/CSS, WebAssembly, Accessibility, UI architecture, Jest, Integration testing
Platforms:	Node.js, Azure, Cosmos DB, WebSockets, API design, LLM integrations, Observability
Leadership:	Cross-team delivery, Technical roadmaps, Architecture documentation, Mentoring, Stakeholder alignment

Professional Experience

GitHub/Microsoft *2020–Present*

Staff Software Engineer—GitHub Copilot

- Led architecture and cross-team delivery of Copilot Remote Control across CLI, web, and mobile; shipped to general availability after a preview reaching 10K daily users and 30K remote sessions per day
- Defined shared session and event contracts for cross-client synchronization, preserving forward and backward compatibility and unblocking mobile and VS Code integrations
- Partnered with product and client teams to set milestones, resolve API requirements, and adjust launch scope in response to user feedback; delivered enterprise policy controls across five repositories
- Leading service architecture refactoring to improve composability and flexibility, alongside real-time communication enhancements using Azure Web PubSub and WebSockets
- Mentored engineers through platform design and ownership handoffs, including session event translation, UI plugins, and secure storage and reauthorization of private data
- Coordinated model-capacity planning across five teams, building a unified usage dashboard and establishing forecasting standards to address rate-limiting risks

Senior Software Engineer—GitHub Repos and Copilot

- Architected the React rewrite of GitHub's file viewing experience, serving 70 million views per day, and established patterns for React applications across GitHub.com
- Designed an extensible frontend plugin architecture adopted by three product teams, enabling independent feature development without cross-team bottlenecks
- Led architecture and phased delivery of an AI workflow platform, including Cosmos DB storage, authentication, observability, and a lossless migration from browser storage to a backend service
- Integrated repository, pull request, and issue context into Copilot Chat and delivered multi-model experiences with OpenAI o1, Google Gemini, and Anthropic Claude
- Established architecture decision records and documentation practices that onboarded seven or more engineers and aligned security and infrastructure partners

Software Engineer III—Azure DevOps

- Owned and optimized the Work Item Query Language and query auto-optimization engine, assuming responsibility for an area previously maintained by an entire team
- Improved Azure Boards accessibility and resolved live-site incidents supporting enterprise customer SLAs

SAS Institute—Data Visualization R&D *2017–2020*

Senior Associate Software Developer

- Built browser applications and libraries for SAS Viya using TypeScript, JavaScript, C++, and WebAssembly, including a customer-facing data visualization template editor
- Led development of an integration testing framework spanning JavaScript host code and a WebAssembly-based C++ runtime
- Drove adoption of shared testing and developer tools across teams through inner-source contributions and support

MetLife, Inc. *2014–2016*

Software Development Engineer II

- Developed an enterprise underwriting application supporting group benefits renewals for over 10,000 companies per month; rebuilt search to reduce end-to-end request time by a factor of 1,000
- Technically led a team building a Spring Boot and Angular proof of concept that became a standard for new projects; advised teams adopting the architecture
- Taught AngularJS and Play to senior engineers and automated more than a dozen recurring manual reports per month

UNC Department of Physics *2013–2014*

Supplemental Instruction Leader

- Led weekly physics problem-solving sessions and authored a 100-page supplemental textbook explaining complex topics

Projects & Leadership

@josh-brown/vector

- Published a TypeScript linear algebra package on npm with matrix factorizations, eigenvalue algorithms, and machine learning applications; achieved 100% function and 99.5% line test coverage

Unicorn CLI

- Built a CLI for SAS's distributed build system, replacing manual steps with an automation-friendly workflow adopted by engineers and contributors across the division

Founder and Music Director, UNC Tarpeggios a cappella group

- Founded a university-recognized ensemble, recruited members, and led over 300 rehearsals and seven full-length concerts; managed production of a $10,000 studio album

Education

University of North Carolina at Chapel Hill

- B.S. in Applied Mathematics with Highest Distinction; B.A. in Physics; Minor in Music
- Chancellor's Carolina Scholarship: four-year full ride; UNC-Chapel Hill National Merit Scholarship
- Phi Beta Kappa; Sigma Pi Sigma; GPA 3.8